

Few-Shot Learning of SERS Chemical-Substrate Pairs: Toward Substrate-Agnostic Sensing

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Abstract

Surface-Enhanced Raman Spectroscopy (SERS) is a powerful tool for chemical identification, yet its practical deployment is often hindered by significant spectral variability across different substrates. Traditional classification models often struggle to generalize across these unique chemical-substrate interfaces, requiring massive libraries that are difficult to maintain. This work utilizes Siamese Neural Networks (SNN) to implement a data-efficient framework designed to identify specific chemical-substrate pairs with limited reference data.

The model employs a contrastive loss-based architecture to learn a high-dimensional embedding space where spectra are clustered by their unique chemical-substrate signatures. Initially validated on standard Raman spectra with 100% accuracy to establish a performance baseline, the system was extended to SERS spectra. While the network’s encoder was developed using a few-shot training strategy ($n = 5$ spectra per class), the system demonstrates true one-shot capability at inference. By comparing unknown queries against only a single known reference spectrum per class, the network achieved a top-1 accuracy of 98.76% on a test set of 403 unseen spectra.

Clustering analysis (t-SNE) of the learned embeddings confirmed that the network provides a geometrically superior representation compared to raw spectra, effectively grouping identical chemical-substrate pairs while distinguishing them from variations in other analyte-surface interactions.

By mastering the representation of the analyte-nanostructure interface, this approach represents a vital progression toward the ultimate goal of a substrate-agnostic chemical identification system. These results demonstrate that Siamese architectures offer a scalable, robust solution for real-world SERS metrology where reference standards are scarce.

Keywords: Surface-Enhanced Raman Spectroscopy (SERS), Siamese Neural Networks, One-Shot Learning, Chemical Identification, Metrology.